

693. Binary Number with Alternating Bits

Solved 

Easy

Topics

Companies

Given a positive integer, check whether it has alternating bits: namely, if two adjacent bits will always have different values.

Example 1:

Input: `n = 5`

Output: `true`

Explanation: The binary representation of 5 is: 101

Example 2:

Input: `n = 7`

Output: `false`

Explanation: The binary representation of 7 is: 111.

Example 3:

Input: `n = 11`

Output: `false`

Explanation: The binary representation of 11 is: 1011.

Constraints:

- $1 \leq n \leq 2^{31} - 1$

Python:

class Solution:

def hasAlternatingBits(self, n: int) -> bool:

`x = n ^ (n >> 1)`

`return (x & (x + 1)) == 0`

JavaScript:

var hasAlternatingBits = function(n) {

 let x = n ^ (n >> 1);

 return (x & (x + 1)) === 0;

};

Java:

class Solution {

 public boolean hasAlternatingBits(int n) {

 int x = n ^ (n >> 1);

```
    return (x & (x + 1)) == 0;  
}  
{
```